Stage 1 – Baseline Diagnostic & Cost-Aware Forecasting

📅 28 February 2026 | 11:00 AM

1. Executive Context

Lumina Energy operates a suburban Mumbai distribution zone.

Under Maharashtra's **Availability Based Tariff (ABT)** regulations, Power Distribution Companies must submit a **2-day ahead load forecast at 15-minute resolution** to the State Load Dispatch Centre (SLDC).

Deviation from submitted schedules attracts financial penalties.

Under ABT:

- Under-forecasting (Actual > Forecast) → costly grid drawal
- Over-forecasting (Forecast > Actual) → inefficiency + deviation charge
- Penalties are asymmetric
- Peak-hour deviations carry higher financial and operational risk

Forecast accuracy is no longer a statistical objective.

It is a **financial exposure minimization problem under regulatory constraint**.

You are engaged as the **Forecast Risk Advisory Team**.

2. Regulatory Simulation Framework

For Stage 1 simulation, assume:

Base Deviation Penalty Structure

Condition	Penalty
Actual > Forecast	₹4 per kWh deviation
Forecast > Actual	₹2 per kWh deviation

Peak Hour Definition

Peak Hours: **6:00 PM – 10:00 PM (local time)**

Peak deviations carry elevated system risk but follow base penalty rates in Stage 1.

3. Data Provided

A. Electric Load Data

- DateTime (15-minute intervals)
- Load (kW)
- Period: 1 April 2013 – 30 April 2021 (Training)

B. Weather Variables (15-minute aligned)

- Temperature
- Heat Index
- Humidity
- Rainfall
- Cool Factor

C. Event Data

- Date
- Event Name
- Holiday Indicator
- Remarks

Historical data includes structural disruptions.
Full stationarity should not be assumed.
External augmentation is permitted but must be documented.

4. Advisory Mandate

You must:

1. Develop a **2-day ahead load forecasting model** (15-min resolution).
2. Incorporate relevant exogenous drivers.
3. Quantify forecast uncertainty (intervals or quantiles).
4. Backtest historical deviation penalty exposure.
5. Propose a cost-minimizing forecast strategy.

Forecasting must minimize **expected financial penalty**, not merely RMSE.

5. Mandatory Outputs

A. Forecast Model Documentation

- Modeling approach
- Feature engineering logic
- Handling of seasonality and structural breaks
- Time-aware validation methodology
- Leakage controls

B. Historical Backtest Metrics

Report on Training period:

- Total deviation penalty (₹)
- Peak-hour penalty (₹)
- Off-peak penalty (₹)
- Forecast bias (%)
- 95th percentile absolute deviation (kW)

C. Risk Strategy Proposal

- Bias positioning (if any)
- Quantile or buffering approach
- Peak-hour risk mitigation strategy
- Expected penalty reduction vs naive baseline

Only genuine data inconsistencies may be raised.

Stage 2 – Regime Shift & Penalty Escalation

📅 28 February 2026 | 7:00 PM

1. Data Release

Dataset now available: **Test Set** (Out-of-Time)

This dataset represents out-of-time load behaviour characterized by:

- Elevated volatility
- Increased intra-day fluctuation
- Altered weekday/weekend elasticity
- Amplified peak-hour variability

This period was not included in training.

2. Regulatory Update – Peak Sensitivity Escalation

Effective immediately:

Revised Peak-Hour Penalty (6:00 PM – 10:00 PM)

Condition	Penalty
Actual > Forecast	₹6 per kWh deviation
Forecast > Actual	₹2 per kWh deviation

Off-peak penalties remain unchanged.

Underestimation during peak hours now carries materially higher financial exposure.

3. Revised Analytical Requirements

You must:

1. Generate forecasts for Test dataset.
2. Compute deviation penalties under revised structure.
3. Quantify:
   - Total penalty (₹)
   - Peak-hour penalty (₹)
   - Off-peak penalty (₹)
   - Forecast bias (%)
   - 95th percentile absolute deviation (kW)
4. Compare:
   - Stage 1 historical penalty exposure
   - Stage 2 test penalty exposure
5. Recalibrate forecasting and buffering strategy if necessary.

4. Modeling Discipline

You must:

- Avoid temporal leakage.
- Ensure weather variables reflect available forecast horizon.
- Document structural adjustments.
- Justify any bias or quantile modifications introduced post-update.

No regulatory clarifications will be provided.